

Northstar Orchard Pilot Report

Fictional Internal Report - August 2026

Executive Summary

Northstar Orchard is a fictional precision-agriculture pilot operated by Arborline Labs in the town of Bellweather. The pilot ran from 6 April 2026 through 28 July 2026 across three orchard blocks: Cedar, Maple, and Juniper. Its goal was to test whether low-cost soil sensors, targeted irrigation, and weekly drone surveys could reduce water use without lowering fruit quality.

The pilot manager was Mira Sen. The field engineering lead was Tomas Vale, and the agronomy reviewer was Dr. Elian Roe. The project used the internal codename ORBIT-17. These names and identifiers are fictional and were created specifically for testing document-question-answering systems.

Key Outcomes

Metric	Result
Total orchard area	18.4 hectares
Water use reduction	21.6% versus the 2025 baseline
Average fruit-size change	+3.2%
Sensor uptime	96.8%
Drone survey cadence	Every 7 days
Pilot budget	INR 18.75 lakh

Block Performance

Cedar block produced the largest water saving at 26.1%. Maple block saved 19.8%, while Juniper saved 18.9%. Juniper had the highest fruit-size improvement at 4.7%, but it also experienced the most sensor maintenance events because two moisture probes failed after heavy rain on 14 June.

The irrigation controller was configured with a soil-moisture trigger of 24%. If a block's median reading fell below that threshold for two consecutive sampling windows, a 22-minute irrigation cycle was scheduled. The exception was Cedar block, where a 19-minute cycle was used because its soil retained water longer.

Operational Findings

The most useful operational change was not the sensor hardware itself, but the combination of sensor readings with drone-derived canopy temperature. During the pilot, the team classified a block as irrigation-priority when median soil moisture was below threshold and canopy temperature exceeded the block's seven-day rolling average by at least 2.4 degrees Celsius.

A false alarm occurred on 3 July in Maple block when a drone image showed an apparent hot zone. Field inspection found that the cause was reflective bird-netting rather than plant stress. After that event, the team added a manual verification step before scheduling any emergency irrigation based only on imagery.

Incident Log

Date	Block	Incident	Resolution
14 Jun 2026	Juniper	Two moisture probes failed after heavy rain	Replaced on 16 Jun
03 Jul 2026	Maple	False canopy-temperature alert	Added manual image verification
19 Jul 2026	Cedar	Pump valve response delayed by 11 minutes	Valve controller rebooted

Cost Notes

The approved pilot budget was INR 18.75 lakh. Final recorded spend was INR 17.92 lakh. The largest categories were drone operations at INR 5.10 lakh, sensors and gateways at INR 4.65 lakh, irrigation-control upgrades at INR 3.20 lakh, and field staffing at INR 2.95 lakh. The remaining amount covered travel, replacement parts, and data storage.

Recommendation

The team recommended extending the system for one additional season before any large-scale deployment. The next phase, internally named ORBIT-18, should include at least 60 hectares, introduce automatic probe-health checks, and compare two irrigation thresholds instead of one.

The review board specifically decided not to automate emergency irrigation in the next phase. Emergency cycles must remain human-approved until the false-positive rate of image-based alerts falls below 2%. The target date for the ORBIT-18 readiness review is 12 February 2027.

Facts Useful for QA Testing

Question	Answer
What was the pilot codename?	ORBIT-17
Who was the pilot manager?	Mira Sen
Which block saved the most water?	Cedar block, 26.1%
What was the soil-moisture trigger?	24%
Why was the Maple alert on 3 July false?	Reflective bird-netting
What was the final spend?	INR 17.92 lakh
What is the next phase called?	ORBIT-18
When is the readiness review?	12 February 2027

Note: This document is entirely fictional and was generated as a test fixture for local LLM and RAG experiments.